

Marketable methods

Education as psychology's primary market

Thus far, we have traced the development of some of psychology's investigative practices in terms of the general knowledge interests pursued by psychologists. This may be adequate for the more general features of these practices but when we turn to more specific features it becomes necessary to take into account a broader social context. The fact is that almost from the beginning of the twentieth century psychology ceased to be a purely academic discipline and began to market its products in the outside world. That meant that the requirements of its potential market were able to influence the direction in which psychology's investigative practices were likely to develop. Practices that were useful in the construction of specific marketable products were likely to receive a boost, whereas practices that lacked this capacity were henceforth placed under a handicap.

Of course, the requirements of the market did not act on a passive discipline. Not only did many psychologists actively court practical application, but these efforts would have led nowhere if the discipline did not have at its disposal certain techniques that lent themselves to the development of a socially useful product. In the present chapter we will analyze two such techniques in terms of their significance for a socially relevant investigative practice: the Galtonian approach to individual differences and the experimental use of treatment groups. Both of these methodological innovations assumed enormous importance in the subsequent development of modern psychology, but it is doubtful whether this would have been the case had they not played a crucial role in the constitution of psychology as a socially relevant discipline.

However, if there is to be any real analysis of the social significance of these techniques, the category of social relevance will have to be brought

down to earth. The abstract ideal of a socially useful science may have occupied an important place in the ideology of many American psychologists. But as soon as they tried to convert this ideal into practice, they had to accommodate themselves to the specific opportunities offered by a particular historical context. In principle the possibilities of applying psychological knowledge might be unlimited, but the actual possibilities available here and now were always sharply circumscribed. They depended on existing institutional forms and on the requirements of those who could command the social resources for putting psychological knowledge to work.

Historically, the one area of social practice with which psychology had established firm links was the field of education. During the nineteenth century the Herbartians had laid the groundwork here, and even though their role in the development of modern psychology was minimal, the expectation that psychology had something to contribute to education had been firmly established. When psychology was transformed from being primarily a system of ideas to something more like a system of practices this expectation was naturally extended to those practices. Not only was this extension made by some psychologists, but it was not an altogether alien suggestion for many educationists. However, the specific nature of the contributions made by modern psychology would depend on the fit between what the psychologists could deliver and what the educationists required.¹

What was it that the educationists required? Given the enormous differences in educational institutions and in the social demands placed on these institutions in different countries, there was considerable national divergence. In the present chapter we will focus on the situation in the United States where the collaboration between educationists and psychologists was particularly effective in the early part of the twentieth century. However, the term "educationists" is almost as vague as the term "social relevance." There were all kinds of educationists, from humble grade school teachers to powerful superintendents of school districts, and their requirements were far from identical.

During the period when the involvement with educational issues played an important role for American psychology there appears to have been a definite shift in the pattern of professional alliances. In the earlier period, characterized by the child study movement of the 1890s, links between psychologists like G. S. Hall and teachers were clearly prominent.² The kinds of investigative practices in which this collaboration was expressed were of the crude, psychological census-taking variety referred to in chapter 5. Nothing more sophisticated would have worked, given the limited training and scientific background of the teachers. This alliance may have done nothing to enhance psychology's status as a science, but it probably helped to popularize the subject and, above all, it translated a vague conceptual

link between psychology and education into a tie between two professionalizing groups.

After the turn of the century psychologists' relations with teachers became increasingly overshadowed by a new professional alliance which was consummated through the medium of a new set of investigative practices. The group of educators with whom psychologists now began to establish an important and mutually beneficial alliance consisted of a new generation of professional educational administrators. This group increasingly took control of a process of educational rationalization that adapted education to the changed social order of corporate industrialism.³ The interests of the new breed of educational administrator had little in common with those of the classroom teacher. Not only were the administrators not directly concerned with the process of classroom teaching, they were actually determined to separate their professional concerns as much as possible from those of the lowly army of frontline teachers.⁴ In this context they emphasized scientific research as a basis for the rationalized educational system of which they were the chief architects. In the United States the needs of educational administration provided the first significant external market for the products of psychological research in the years immediately preceding World War I.

But to the potential clients of the academic psychologists research meant something that was rather different from the latter's traditional laboratory practice. Research, to the administrators, was an activity whose results had to be relevant to managerial concerns. It had to provide data that were useful in making immediate decisions in restricted administrative contexts. This meant research that yielded comparable quantitative data on the performance of large numbers of individuals under restricted conditions. Excluded was research that went beyond the given human and social parameters within which the administrators had to make their decisions. It was, in other words, technological research that would help in dealing with circumscribed problems defined by currently unquestioned social conditions. Not infrequently, administrators simply needed research for public relations purposes, to justify practices and decisions they judged to be expedient.

In representing the problematic that defined their goals these early educational administrators very frequently made use of an industrial metaphor. As one of them put it in an unforgettable image, sustained over several pages, "education is a shaping process as much as the manufacture of steel rails."⁵ Elaborate analogies were drawn between stockholders and parents, general managers of factories and superintendents of schools, foremen and principals as well as industrial workers and teachers.⁶ Within this framework nothing was more natural than the extension of the principles of "scientific management" from industry to education. In the con-

siderable literature devoted to this topic, the most succinct formulation of these principles, as they were understood by the educational administrators, appears to have been given by Frank Spaulding in 1913:⁷

1. The measurement and comparison of comparable results.
2. The analysis and comparison of the conditions under which given results were secured – especially of the means and time employed in securing given results.
3. The consistent adoption and use of those means that justify themselves most fully by their results, abandoning those that fail so to justify themselves.

Such a scheme assigned an important and clearly defined role to research. It had to provide the information necessary for steps one and two, so that the administrator could take the appropriate action specified by step three. Two kinds of research data were required: first, comparable measurements of results, defined as individual performance; and second, a comparison of the relative efficiency of various conditions in producing these results.⁸ What was needed were scales that measured performance and experiments that assessed the relative effectiveness of such conditions as were potentially under the control of the administrators who had to choose between them. The primary consumers of the results of such research were those whose controlling position made the results potentially useful to them, not those who participated in the research as subjects.

A significant number of psychologists quickly responded to the requirements of a type of research that was quite different both from the kind of laboratory practice that had so far defined scientific psychology and from the computations of child study. By 1910 this new kind of research practice had gained sufficient momentum to lead to the publication of a separate journal devoted to it, the *Journal of Educational Psychology*. The core of contributors was formed by those, like Thorndike at Columbia Teachers College, whose links with educational administrators had already been firmly cemented.⁹ But even fairly traditional experimentalists, like Carl Seashore and Edmund Sanford, found the siren call of the new applied psychology immediately appealing. “So long as the science must be financed by appropriations and endowments,” Sanford wrote in the first volume of the new journal, “it [the new applied psychology] furnishes an effective defense against ignorant and hostile criticism and a tangible excuse for the investment of institutional capital.”¹⁰

What Sanford did not anticipate was the fact that the attempt to apply psychology to the requirements of educational administration would have consequences for psychology that were far more profound than any contribution psychology was likely to make to education. In the first place, it meant a decisive break with the kind of educational psychology that had been envisaged by the pioneering giants of American psychology, by James,

Baldwin, Hall, and Dewey. Their broader vision was now replaced by a much narrower and purely instrumental conception of what psychology could accomplish.¹¹ The institutional constraints that the new educational psychologists took for granted required them to emphasize the passivity of the child and to restrict themselves to measured performance rather than wasting precious resources on an exploration of mental processes that had no obvious utility in terms of the goal of choosing the conditions that were most efficient in producing predetermined results.

The new pattern of psychological research practice that originally crystallized in the work of educational psychologists proved to be increasingly attractive to other psychologists in the years that followed. The experience gained in research for educational administration made it possible for psychologists to exploit the professional opportunities presented by the large-scale requirements of military administration in World War I. Without the methods of mental measurement recently tried out within an educational context, American psychologists would have had nothing immediately useful to offer the military authorities.¹² It was at this time that the entire field of applied psychology came to be defined in terms of psychological knowledge that would be useful in administrative contexts. During the 1920s and 1930s the conviction that psychological research had to produce the kind of knowledge that would be potentially useful in certain practical contexts characterized an important and growing section of academic psychologists. For people like Woodworth, psychology's affiliation with education had replaced its earlier affiliation with philosophy.¹³ But what this concern with applicable knowledge generally came down to in practical terms was the desire for knowledge that was marketable in administrative contexts. This necessarily entailed a growing preference for a type of research practice that was capable of yielding the kind of data that could be perceived as relevant to the interests of administrators.

Close links between social administration and psychological practice entailed significant effects for both. Certainly, psychological practice had social repercussions, and at least some aspects of this relationship have received a great deal of attention in the past.¹⁴ But of interest in the present context is the reciprocal influence of social practice on psychological investigation. That such an influence existed was clear to central figures like Thorndike, who was certain that "the science of education can and will itself contribute abundantly to psychology."¹⁵ More particularly, the demands of a certain kind of market greatly favored specific research practices that could produce the kind of knowledge that was usable in this context.

In the broadest terms, the kind of knowledge that was most obviously useful in administrative contexts was statistical knowledge. In such contexts information about persons was generally of interest only insofar as it pertained to the categorization of individuals in terms of group characteristics. Dealing with individuals by categories constitutes the essence of admin-

istrative practice. One general consequence of the impact of administrative concerns on psychological research has already been described in chapter 5: Psychologists increasingly turned away from an individual attribution of their findings to an attribution that worked with statistically constituted collective subjects. It is not surprising that this development characterized the applied-research literature from its beginnings. Applied research generally aimed to supply administrators with the kind of knowledge that they would need to make efficient decisions. Only quantitative statistical knowledge could serve this purpose in most practical situations.

But it is possible to go further than this global characterization of the new psychological knowledge products. The requirements of the more articulate administrators were in fact more specific, as we have seen. They wanted methods for the comparison of conditions and the measurement of results. Psychologists obliged by concentrating on certain incipient aspects of their own practice that could be directly adapted to these requirements. The result was that these aspects attained an extraordinary prominence in psychological research practice. What were these aspects?

In a strategic article, published in an early volume of the *Journal of Educational Psychology*, E. C. Sanford, a distinguished researcher who was by this time president of Clark College, identified three basic methods that psychologists would be contributing to education as a rationalized social practice:¹⁶ First, there is standard laboratory work, as exemplified by German investigations of “economic methods of memorizing”; the pedagogical implications of this work had been explored by Meumann, but the basis of the method had been established by Ebbinghaus. Second, there was the work on individual differences as exemplified by the method of mental testing that was just then getting into its stride. Third, there was the classroom experiment, whose major pioneer was W. H. Winch in England; in this type of investigative practice groups of school children, matched in ability, would be exposed to different methods of instruction and their performance before and after the experimental intervention would be assessed. The first and third methods were dedicated to comparing the efficiency of different techniques of learning and instruction; the second method was used to select individuals for programs rather than the other way around. All three methods were nicely in tune with the stated requirements of educational administration.

In examining the kind of investigative practice that was promoted by psychology's reliance on a particular market for its research products, we will begin with the second and third of Sanford's methods, mental tests for individual differences, and classroom experiments. These depended very directly on the educational market. The study of the economy of memorizing had more solid roots in certain practices of laboratory psychology, which will be examined in chapter 9.

Mental testing as an investigative practice

The too-ready marketability of mental tests profoundly affected the shape of psychological research. In the period after World War I there was an enormous expansion of work based on the use of mental tests. This work had so little connection with the more traditional forms of psychological experimentation that it eventually seemed as though there were two distinct disciplines of scientific psychology, one experimental and the other based on the correlational statistics used in mental-test research.¹⁷

Mental testing flourished because of an interest in individual differences, but this observation hides more than it reveals. The term "individual differences" taken in isolation from a specific context is exceedingly vague and could just as easily apply to the work of the novelist as to that of the psychologist. Clearly, when mental testing is derived from an interest in individual differences, it is not this very general meaning that is relevant but a very specific meaning that is conveyed by the context. It was an interest in looking at individual differences in a particular way that found expression in the development of mental tests.

Indeed, the investigation of individual differences preceded the development of modern mental testing by many years. There were old interpretive practices of reading an individual's character with the help of bodily signs. These might be based on somatic indications, as in the classical doctrine of temperaments, or on facial characteristics, as in the relatively more recent versions of physiognomy.¹⁸ During the first half of the nineteenth century the practice of phrenology built on, and to a large extent replaced, these older practices. Somewhat later, the measurement of the brains of various kinds of people and the physiognomic characterization of criminals¹⁹ followed on in the same tradition. Gradually, measuring practices came to replace the older interpretive practices, reflecting a changing conception of the nature of scientific knowledge. What distinguished mental testing from these older approaches was neither an interest in the assessment of individual differences nor a preference for quantitative methods as such. The difference resided rather in the medium through which individual differences were assessed. Mental tests severed the very ancient link between psyche and soma and proposed to assess individual differences entirely by measuring function. In practice, this meant measuring individual performance in restricted situations.

Assessing individual differences in terms of specific performance measures, while relegating any other method of assessment to the realm of the unscientific, amounted to a redefinition of the object of investigation. Psychological differences were now defined in terms of differences on measures of individual performance. It was performance at uniform set tasks that counted psychologically, not, for example, facial expression or artistic style.

Even on this somewhat restricted basis a broadly conceived study of individual differences was entirely possible. In fact, during the early modern period in the investigation of individual differences most investigators were not interested in the comparison of individual performances as such but in the characterization of psychological types and of human individuality. Before the advent of intelligence tests, Alfred Binet's name was associated with an "individual psychology" that was truly a psychology of individuals, in the sense that psychological performance measures were used to assess the individual style of a person's functioning.²⁰ Another European authority in this area, William Stern, distinguished between the study of human variety and the study of individuality, and he was inclined to accord a higher value to the latter.²¹ James Mark Baldwin had criticized the Wundtian style of psychological experimentation for treating individual differences as a nuisance, rather than as an object of study. But he conceived of individual differences in typological terms, as was general at the time.²²

What the development of mental testing did was to redefine the problem of individual differences in terms of a *comparison* of performances. The quality of a performance could no longer be used for describing individuality or for analyzing typological patterns. Instead, a measure of individual performance was to be used for specifying the individual's position with respect to an aggregate of individuals. Comparing individual performance to a group norm, however, meant that the characterization of any individual depended just as much on the performance of a set of others as it did on anything he did himself. The whole exercise depended on the assumption that the salient qualities for characterizing an individual were qualities that he or she shared with others rather than qualities that were unique.²³ These common qualities had to be thought of as constant elements whose nature was unaffected by their cohabitation with other such elements in the same individual. Carried to its logical conclusion, this methodology for assessing "individual differences" actually eliminated individuals by reducing them to the abstraction of a collection of points in a set of aggregates.²⁴

Although the statistics of individual differences constituted the very antithesis of an interest in psychological individuality, they were able to speak quite directly to another kind of concern: the problem of conformity. The new psychological practice was based on the setting up of "norms" in terms of which individuals could be assessed. In most cases these norms were psychological only by inference; in the first place they were norms of social performance.²⁵ The categories in terms of which individuals were graded were not generally socially neutral categories but carried a powerful evaluative component. It was not an academic interest in the psychology of human cognition that motivated the normative study of individual performance but an interest in establishing who would most effectively conform to certain socially established criteria. These criteria ranged all the way

from the unidimensional “general intelligence” of the eugenicists to the qualities needed in a good salesman. But they were always criteria that only made sense in the context of particular social interests, be they grand and ideological or practical and mundane.

Throughout most of the nineteenth century, the concern with individual differences had been fueled by two different sets of interests, one private, the other public. In a competitive society, offering certain possibilities for individual social mobility, individuals had an interest in assessing their personal potential so as to make good use of any opportunities for social advancement that might present themselves. For several decades phrenologists were able to exploit this as a market in aids to self-improvement,²⁶ and to some degree psychologists were later able to do likewise. In the course of time, however, a much more important market opened up for experts on individual differences. Those who administered social programs and institutions always had a potential interest in such expertise, insofar as it could assist in the administrative process – specifically by providing a culturally acceptable rationale for the treatment of individuals by categories that bureaucratic structures demanded.

With the enormous increase in the size and scope of bureaucratically administered institutions during the latter part of the nineteenth century, especially in the fields of education and the management of social deviance, the need for special sorting practices increased greatly. In fact, social sorting on apparently rational criteria became the major function of these institutions. At first, the task was accomplished by adapting a social practice that had recently become popular in a related context, namely, the *examination*. Two kinds of examination had existed for a long time, the academic and the medical. By the middle of the nineteenth century both had been adapted to new tasks. The academic examination had been adapted to the selection of senior administrators, and the medical examination had been transformed into the psychiatric examination, which, among other things, assessed moral fitness.²⁷ In the latter part of the century the academic examination was further modified to make it suitable for the mass selection and grading of ever larger sections of the population, while the distinction between intellectual and moral worth was becoming blurred. The advent of mental tests represented a continuation and culmination of this development. Finally, the notion of a unitary, biologically fixed “intelligence” provided a license for grading the entire population as though they were members of one school class.

Those who were in control of the social institutions in which mental tests found their natural home were looking for ways to sort individuals in terms of their measuring up to institutionally appropriate criteria. They needed effective techniques for making selection decisions that would enhance the efficiency of existing institutional structures. The characterization of individuals in terms of their degree of conformity to criteria of performance

within specific social contexts made it easy to redefine the problems of institutions, or even the problems of society, as individual problems. If all social problems were nothing more than the aggregate of individual problems, they could be handled by appropriate treatment of individuals and required no questioning of the social order.

Within psychology the growing popularity of research based on one or other form of mental testing accounted for a considerable part of the expanded use of "collective subjects," which we noted in chapter 5. This involved the large-scale appearance of both "natural" and psychometrically defined groups in psychological research reports. No longer did psychological research mean the experimental exploration of individual minds. The individual was of interest only in terms of his or her standing in an aggregate. Research objectives largely shifted to the comparison of such aggregates and the statistical relationships between them. In certain practical settings ("psychological clinics"), a more individualized employment of mental tests, closer to the original vision of Binet, did continue.²⁸ But insofar as such practice claimed to have a scientific basis, that basis was also statistical. Even though individual patterns might be considered, they were still patterns of performance defined in terms of common group norms. Attempts to relate clinical practice to an individualized experimental psychology were unsuccessful.²⁹ What survived were packets of individualized clinical practice whose claims to scientific expertise depended on their reliance on statistical research data. Practically relevant research practice was generally devoted to work with statistically constituted psychological objects.

Marketable mental testing ensured a dominant place in psychological research for the style of investigative practice that had been pioneered by Galton. This applied to the social construction of the investigative situation where some of the features of Galtonian anthropometry were exaggerated to the point of bizarre caricature in the mass-testing methods developed in military, but also in educational, contexts.³⁰ But it also applied to the way in which statistical techniques were used to create the objects that were the real focus of the researchers' scrutiny. These techniques no longer functioned to make the observations of individuals more precise but came to play a crucial role in constituting the objects around which research revolved – statistical distributions of scores contributed by a mass of individuals.³¹

The ready adoption of the Galtonian model by a significant section of American psychology is not at all surprising when one takes into account certain parallels between the situation faced by Galton and that which confronted early twentieth-century American psychologists. Galton wanted to establish a science of human heredity although it would be impossible to ground such a science in anything resembling controlled experimentation. His alternative was to base this science on the statistical comparison

of group attributes.³² Many American psychologists were interested in promoting a socially relevant science involving aspects of human conduct and performance that were not accessible to precise experimental study under controlled conditions. The emerging science of biometrics offered statistical techniques, initiated by Galton and developed by Pearson, which promised to provide ways around this problem. Something that looked like a science could apparently be created by statistical rather than experimental means.

There was however a residual problem about establishing psychological knowledge claims on a Galtonian basis. In the classical psychological experiment, devoted to the exploration of functional relationships (see chapter 2), there was always a fundamental asymmetry that provided the basis for causal attributions. One of the variables was under the experimenter's control and could be "arbitrarily" (Wundt) varied. If this led to regular changes in the response measure, one had grounds for making causal attributions. The situation was quite different in the Galtonian model. Here the investigator attempted to impute relationships in situations over which he had no control. Such relationships had therefore to be symmetrical, because the investigator was unable to influence the situation so as to give the status of cause to one of the variables and the status of effect to the other. What he could hope for was a measure of concomitance or covariation. The descriptive statistics of the Galton-Pearson school³³ provided him with such measures. Such statistics were therefore constitutive of the relationship described, whereas the calculus of error of the Wundtian model could not be said to constitute the causal relationships that were investigated. In the Galtonian model the establishment of statistical correlations among attributes of natural populations became the goal of research, whereas in the classical model statistics had been at best a means for ensuring the reliability of certain observations.³⁴

The adoption of statistical usage in the tradition of Galton and Pearson meant that American psychology was now committed to making two very different kinds of knowledge claim, both of which had serious grounds for being taken as scientific. At first, this produced considerable confusion and a bewildering variety of rationalizations within the discipline.³⁵ Gradually, however, a kind of official doctrine emerged. This established the ideal form of research practice as one that combined the manipulative aspects of experimental procedure with statistically constituted objects of investigation. It was not noticed that this transformation deprived the experimental manipulation of its original rationale, which was the production of some causal process in an individual psychophysical system. Interest had shifted to the statistical outcome of experimental trials as manifested in differences between individuals.³⁶

This was quite acceptable to the growing number of American psychologists who saw their scientific work as devoted to the development of

techniques for the social control of individuals. Such a psychology had to be able to use its measurements to make predictions about future effects that could be taken into account by those in controlling positions. In principle such predictions about future effects, reliably to be expected on the basis of previous measurements, could of course be made on a purely statistical basis. But that would have required really large-scale research with a very extended time frame. Psychology would have to be far better established before it could hope to muster the resources for such research. Even then, the social organization of research and of individual career patterns greatly favored small-scale fractionated research with a relatively brief time span.

It is therefore not surprising that psychological research in the Galtonian style did not establish its claim to social relevance on the basis of sophisticated long-term statistical studies. There was a highly effective shortcut available to it – to hitch its wagon to prevailing preconceptions regarding causation in human affairs. Those preconceptions prescribed that human interaction was to be interpreted as an effect of stable, inherent causal factors characterizing each separate individual. The most important of these factors were defined as psychological in nature, intelligence, and temperament to begin with. Such factors were causal in the sense that they set rigid limits for individual action and were themselves unalterable. The link to hereditarian dogma was quite direct, and many of the major proponents of the Galtonian style of research in psychology were also keen eugenicists.³⁷ The statistical patterns they constructed were interpreted in terms of hereditary individual traits. For example, a normal distribution would be sought after because of a prior belief that this was the characteristic distribution of fixed biological traits. Psychological knowledge claims in the Galtonian mold generally derived their significance from their reference to inherent properties that were built into each individual, although they could only be assessed by assuming them to be of essentially the same nature in all individuals. The structure of such claims remained unaltered where the original geneticist basis was dropped and replaced by a non-committal reference to “traits.”

The transformation of the old psychology of group differences by Galtonian perspectives led to an important development in the nature of the groups that were the carriers of psychological attributes. Originally, psychologists had simply adopted the categories that defined group membership from everyday life, age and sex being the prototypical examples. But the Galtonian use of statistics greatly facilitated the artificial creation of new groups whose defining characteristic was based on performance on some psychological instrument, most commonly an intelligence test. A score on a mental test conferred membership in an abstract collectivity created for the purposes of psychological research. This opened up untold vistas for such research because psychologists could create these kinds of

collectivity ad infinitum and then explore the statistical relationships between them.

Origins of the treatment group

In chapter 5 we noted that in the first half of the twentieth century the investigative practice of American psychology came to rely increasingly on the construction of "collective subjects" for the generation of its knowledge products. Such constructed collectivities were not to be found outside psychological practice but depended for their existence on the intervention of psychologist-investigators. Three types of these artificial collectivities were distinguished: those that were the result of averaging the performance of individuals subjected to similar experimental conditions, those that were constituted from scores obtained by means of some psychometric testing instrument, and those that were produced by subjecting groups of individuals to different treatment conditions.

Whereas the first type of aggregate represented an outgrowth of traditional experimental procedures, the second type clearly owed its existence to the demands placed on psychological investigative practice by its one major external market, the education industry. However, this market also appears to have been responsible for the emergence of the third type of constructed collectivity, the treatment group.

We have seen that what the educational efficiency experts expected from psychological research was, first, methods of measurement that would permit a comparison of results, and second, a rationalized procedure for assessing the effects of various kinds of administrative intervention. Mental tests satisfactorily fulfilled the first expectation, but what about the second? In order to assess the relative efficiency of alternative educational programs, a way of comparing groups exposed to these programs was needed. Yet traditional experimental psychology, because of its focus on the individual, was of little help in this matter.³⁸ Educational researchers therefore began to find their own ways of coping with the problem.³⁹

The pioneer was W. H. Winch, an English school inspector, who was interested in comparing the relative effectiveness of various classroom conditions on such matters as mental fatigue and transfer of training.⁴⁰ To do this, he subjected equivalent groups of children to different conditions, taking the relevant measurements before and after the intervention.⁴¹ His example was quickly followed by others investigating such topics as the relative merits of different methods of instruction or the effects of speed on learning.⁴²

There was nothing particularly surprising about this development. This was a period during which there was a pervasive tendency to see school problems as problems of work efficiency, a point to which we shall return. Schools were under pressure to work as efficiently as possible. Because

school work was carried out in groups (i.e., school classes), the assessment of its efficiency depended on aggregate measures. In this framework the work of the individual child was of interest only insofar as it contributed to or detracted from the overall effectiveness of the work unit.

What is interesting about these early classroom studies is that they took place in an institutional environment that allowed for a certain systematic manipulation of conditions. Thus, studies of work efficiency readily took on an experimental form. The possibility of investigating the effect that variations in the conditions of class work had on measures of output provided a strong practical incentive for combining the use of group data with the experimental method. It was not long before the first examples of studies using the treatment group approach began to appear in journals devoted to basic research.⁴³ Although such studies were very similar, both in form and content, to the studies conducted in schools, their authors attributed a much broader significance to their results. It was not just the effect of specific classroom conditions on work at school that was now supposedly studied but “fatigue” in general and “learning” in the abstract.

The subsequent history of treatment group methodology ran a rather different course in “applied” psychological research, conducted in institutional settings, and in laboratory research. In the former, there was an initial period of enthusiasm. Within a decade of Winch’s first publication the procedure was being sold to American school superintendents as the “control experiment” and touted as a key element in comparing the “efficiency” of various administrative measures.⁴⁴ Five years later, W. A. McCall, a longtime associate of Thorndike at Teachers College, Columbia, gave the method its textbook codification for educational research.⁴⁵ McCall’s justification of experimentation in schools was strictly financial: “Experimentation pays in terms of cash.” In one illustrative case he estimated “the value of the increased abilities secured” at ten thousand dollars.⁴⁶ Technically, McCall’s text was quite sophisticated for its age, expounding on randomization and complex experimental designs two years before the publication of R. A. Fisher’s much better-known text.⁴⁷

In spite of these promising beginnings, the history of treatment-group methodology in research in education and educational psychology was not exactly a march of triumph. In the *Journal of Educational Psychology* the use of groups constructed along these lines was confined to 14 to 18 percent of empirical articles during all three periods sampled for the present study, 1914–1916, 1924–1926, and 1934–1936. During the later periods experimental studies in the field of education would be more likely to be published in new journals, like the *Journal of Experimental Education*. In due course, however, a certain pessimism about the prospects of experimental research in education began to set in. The claims made on behalf of the quantitative and experimental method had undoubtedly been wildly unrealistic, and in

the light of changing priorities the illusions of the early years were unable to survive.⁴⁸

In the *Journal of Applied Psychology* the use of treatment groups was even less impressive, advancing from 1 to 6 percent of empirical studies between 1917–1920 and 1934–1936. Clearly, there was almost no demand for this type of research practice outside the educational system, and in the latter the demand was limited. This outcome suggests the operation of institutional constraints that tended to impose a limitation on experimental research in institutional settings.⁴⁹ Such research had to be carried out with the approval of authorities who presided over power structures that were for the most part highly bureaucratized. The shape of such research was always at the mercy of institutional priorities and shifts in these priorities. In the field of education the introduction of comparative experimentation coincided with the heyday of the “cult of efficiency,” but once the influence of this movement waned the impetus for the continued expansion of that kind of research was gone.

More generally, there are always limitations on the utilization of an experimental approach in the context of authoritatively structured institutions geared to practical goals and interests. Not only can experimentation constitute a practical nuisance, but more important, it easily becomes seen as a threat to vested interests and the traditional practices of established power groups. Although psychologists could help both with the selection of individuals for preestablished programs and with the selection of the programs themselves, the former function was far safer in that it lacked any potential for competition with the power of established authority to dispose over institutional conditions.

The situation was quite different in the area of laboratory research where experimentation could safely be used as the preferred method. In the safety of university laboratories a kind of experimentation that had originated in relatively mundane practical concerns developed into a vehicle for maintaining fantasies of an omnipotent science of human control. In their original school settings experiments based on treatment groups could be of some practical value because the institutional context provided a known and consistent framework for applying the results. But the more ambitious American psychologists were not content with the humble role of psychological technician. To them the performance of children on school tasks under different conditions was simply one instance of the operation of generalized “laws of learning” that manifested themselves in the entire gamut of human behavior. This faith enabled them to continue to use a style of experimentation that had already proved its practical usefulness and to reinterpret the results of such experimentation so they now provided evidence bearing on the nature of the fundamental laws of behavior.⁵⁰ They imagined that what they observed in these experiments were not specific

effects depending on a host of special conditions but the operation of a few highly abstract principles of organismic learning or forgetting.

The relationship of such a science to practical efforts at optimizing human performance was seen very much like the relationship between physics and engineering.⁵¹ Any practical success at increasing the efficiency of school learning, for instance, was thought to depend on the operation of general laws of learning that existed independently of any context. The abstract laws that psychology sought were conceived to be analogous to physical laws. Physics, and especially astronomy, had had to confront the clash between faith in the generality of physical laws and the undeniable variability of observations. That problem was solved to everyone's satisfaction by showing that errors of observation obeyed statistical regularities. In the case of abstract psychological laws there was the additional problem that these laws would be instantiated in the variable behavior of individuals, human or otherwise. It was widely assumed that that source of variability could be reduced to statistical order by an analogous application of the calculus of error. This assumption allowed psychologists to retain their faith in the existence of abstract, universally valid, laws of psychology in the face of daily confrontations with human and animal variability in investigative situations.

Unquestioned faith in the existence of abstract underlying laws produced an interpretation of inconsistent observations in terms of an assumed continuous variability of individual behavior on a set of common dimensions. This interpretation, however, blurred the distinction between generalizations that expressed a truly general relationship and generalizations that were based on no more than a statistical construction superimposed on heterogeneous material. If this distinction could be avoided – and it certainly was avoided⁵² – one could turn the variability of one's material into an advantage. If one worked with groups of subjects, one could always structure the outcome in terms of some overall central tendency and some measure of individual deviation from that tendency. This possibility made it terribly easy to produce generalizations that could be held to express psychological "laws," provided one operated in a scientific culture that made no distinction between statistical and psychological reality. The rising popularity of group experiments among practitioners of "basic" psychological research was no doubt promoted by this feature.

The basic science that was sought was really an abstract form of the practice of controlling the performance of large numbers of individuals. Its fundamental laws were to take the form of relationships between environmental interventions and changes in the average response to such interventions. For such a science the use of treatment groups in practical contexts would have an exemplary quality. Transferring their use to the laboratory allows one to construct a model of the kind of world that the laws of this new science presuppose. It is a world in which individuals are

stripped of their identity and their historical existence to become vehicles for the operation of totally abstract laws of behavior.

The treatment group was uniquely appropriate for establishing the knowledge claims of a science constructed out of relationships between abstract external influences and equally abstract organisms. Taking the same group of subjects through the same set of environmental variations could also be used for this purpose, and indeed it was. But this approach provided subjects with a bit of history, even though it was only a specially constructed experimental history. This meant that the results could not be unambiguously interpreted in terms of environmental effects on ahistorical subjects.⁵³ Only the treatment group could provide a practical construction that was a pure reflection of the kind of theoretical construction that the so-called science of behavior favored. As the use of this foundation for knowledge claims became more widespread, the empirical basis for questioning the prevailing shape of psychological theory became narrower. In terms of their overall structure experiments increasingly resembled a practical enactment of what theory presupposed.

8

Investigative practice as a professional project

Interpreting the professional project

The previous chapters have provided abundant evidence for the fact that the historical development of psychological research practice did not proceed along a single track. At the very least, there were two different lines of development. By the 1920s the Wundtian style of experimentation, with its roots deep in the philosophical and scientific traditions of the nineteenth-century German university, seemed to constitute a deteriorating research program within American psychology. As it was fast losing its appeal for all but a few practitioners, the Galtonian research program, strongly linked to practical rather than academic concerns from the beginning, was moving from strength to strength. Not only was it extending its appeal with every passing year, but, as we have seen, it proved itself capable of generating exciting methodological innovations that promised to extend the scope of scientific psychology far beyond what had hitherto been thought possible.

We have also seen that the galloping success of the Galtonian program was in no small measure due to its very direct link to the demands of a significant extradisciplinary market for its products. Nevertheless, it would be a mistake to see the development of a certain style of investigative practice as an essentially passive adaptation to external social requirements. That might have been true of certain individuals, but it was not true of the response of the discipline as a whole. Even on an individual level most practitioners recognized that practical successes would be short lived if they threatened to undermine the prestige of the disciplinary enterprise they represented. If the practice of mental testing, for example, proved to be transparently unreliable, it would do the practitioners no good in the long run, no matter how spectacular the immediate yield might appear to be. The practical effectiveness of psychologists depended quite crucially on